

Diagnostic Accuracy of EyeArt for Fundus-Based Detection of Diabetic Retinopathy: A Systematic Review and Meta-analysis



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• **PURPOSE:** Diabetic retinopathy (DR) is a leading cause of preventable blindness globally. Although early detection via routine retinal screening significantly reduces vision loss, screening rates remain suboptimal due to workforce shortages and limited accessibility. Autonomous artificial intelligence (AI) systems such as EyeArt offer an FDA-authorized solution for point-of-care DR screening without ophthalmologist oversight

• **METHODS:** We conducted a systematic review and meta-analysis following PRISMA-DTA guidelines to assess the diagnostic accuracy of EyeArt in detecting referable diabetic retinopathy (rDR) from color fundus photographs. Searches of PubMed, Embase, and ClinicalTrials.gov through April 2025 identified eligible studies involving adult populations screened with EyeArt. Sensitivity and specificity were pooled using bivariate random-effects models. Subgroup and applicability analyses were conducted to evaluate heterogeneity and clinical relevance.

• **RESULTS:** Seventeen studies comprising 162,695 examinations were included. EyeArt demonstrated a pooled sensitivity of 95% (95% CI: 92%-97%) and specificity of 81% (95% CI: 74%-87%). Subgroup analyses indicated consistent accuracy across study designs, economic settings, healthcare contexts, device types, external validation, and image gradability. Specificity varied slightly with vendor involvement.

• **CONCLUSIONS:** Across 17 real-world studies (162,695 examinations), EyeArt exhibits high diagnostic accuracy for detecting referable diabetic retinopathy (pooled sen-

sitivity 95%, specificity 81%), with high certainty for sensitivity and moderate certainty for specificity. Its consistently strong sensitivity supports autonomous screening in primary care. However, variability in specificity—along with inconsistent reporting/handling of ungradable images—warrants attention and standardized quality-assurance. Successful deployment will depend on workflow/EHR integration, sustainable reimbursement, and targeted implementation in underserved populations to maximize public-health impact. (Am J Ophthalmol 2026;281: 465–479. © 2025 Elsevier Inc. All rights are reserved, including those for text and data mining, AI training, and similar technologies.)

INTRODUCTION

DIABETIC RETINOPATHY (DR) IS THE LEADING cause of avoidable blindness in working-aged people worldwide and is becoming more prevalent with rising diabetes patterns.¹⁻³ Around the world, 1.07 million people were blind and 3.3 million had moderate-to-severe visual impairment due to DR in 2020, numbers which are predicted to increase significantly by 2045.^{2,3} Early detection of DR through regular dilated retinal exams reduces the risk of severe vision loss,⁴ but less than half of diabetics obtain recommended screening because of limited healthcare capacity, patient-related barriers, and lack of ophthalmologists.^{5,6} Tele-ophthalmology programs have extended the reach of screening by analyzing retinal images taken in primary care remotely. However, such programs are constrained by the availability of trained specialists, thereby generating interest in automated approaches.⁷

Great strides have been made in both computer-aided DR detection by artificial intelligence (AI) and the use of deep learning—a subfield of AI—now a powerful DR screening modality, with diagnostic accuracy and performance approaching those of expert human graders.^{8,9} Key studies such as Gulshan et al. (2016) presented AI methods with a sensitivity and specificity of more than 90%, comparable or even better than that of human performance.⁸ Follow-up clinical assessments supported these data, includ-

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